

LAB GUIDELINES

Lab 29 — Troubleshooting Common Hardware Failures

Maps to CompTIA Server+ objective **4.2 — Given a scenario, troubleshoot common hardware failures** (predictive failures; memory errors — BSOD/PSOD/dump; utilisation; POST errors; lockups; kernel panic; CMOS battery failure; system lockups; random crashes; fault/device indication — visual LED/LCD/auditory/olfactory; POST codes; misallocated virtual resources; causes: PSU fault, fans, heat sink, seated cards, incompatibility, cooling failures, backplane failure, firmware incompatibility, CPU/GPU overheating; environmental — dust, humidity, temperature; tools — event logs, firmware upgrades/downgrades, hardware diagnostics, compressed air, ESD equipment, reseating cables).

Run all commands on <https://killercoda.com/playgrounds/scenario/ubuntu>

Required software (free, apt): dmidecode, lshw, smartmontools, lm-sensors, memtester, stress-ng, edac-utils

```
apt update && apt install -y dmidecode lshw smartmontools lm-sensors memtester stress-ng edac-utils
```

Step 1 — Event logs first (the rule the exam expects)

```
dmesg -T | tail -40
journalctl -p err -b 0 --no-pager | tail -40
journalctl -k | grep -iE 'mce|edac|temperature|thrott|over.?heat|ecc' | tail
```

Hardware faults almost always print to **dmesg / journal** before the service notices.

Step 2 — CPU thermal + utilisation

```
sensors 2>/dev/null | head # may be empty in a VM
stress-ng --cpu $(nproc) --timeout 10s --metrics-brief 2>&1 | tail
mpstat 1 3
```

Server+ 4.2 cause: **CPU/GPU overheating**. Symptoms: thermal throttling, random lockups, kernel panic. Field response: check **fans, heat sink seating**, dust accumulation; **compressed air, reseal cables**.

Step 3 — Memory: errors, ECC, dumps

```
edac-util -v 2>/dev/null || echo "no EDAC kernel module exposed in this VM"
dmesg | grep -i mce | tail
journalctl -k | grep -i "memory" | tail
# memtester targets specific memory ranges – example syntax:
echo "memtester 100M 1 # would test 100 MiB, 1 pass"
```

Real hosts surface **uncorrectable ECC errors** in EDAC and in the BMC SEL log (`ipmitool sel list`). Two corrected errors on the same DIMM = swap that DIMM.

Crash dump terms the exam tests:

- ****BSOD**** (Blue Screen — Windows)
- ****PSOD**** (Purple Screen — VMware ESXi)
- ****Kernel panic**** (Linux)
- ****Memory dump**** = `/var/crash/`` (Linux, via `kdump`) or `MEMORY.DMP`` (Windows)

Step 4 — Storage health (predictive failures via SMART)

```
smartctl -a /dev/sda 2>/dev/null | head -40
smartctl -H /dev/sda 2>/dev/null
smartctl -l error /dev/sda 2>/dev/null | head -20
```

SMART attributes that **predict** drive failure:

- ``Reallocated_Sector_Ct``

- `Current\_Pending\_Sector`
- `Offline\_Uncorrectable`
- `Wear\_Leveling\_Count` (SSD)
- `Media\_Wearout\_Indicator` (SSD)

When any climbs above 0 → **schedule replacement** before it fails.

Step 5 — PSU and fan faults via IPMI

```
# On real hardware:
# ipmitool sensor list | grep -Ei "fan|psu|temp|voltage"
# ipmitool sel list | head
echo "BMC SEL captures PSU input loss, fan stall, temp threshold crossings"
```

Symptoms of PSU fault: random power-off, system instability, kernel `power_meter` warnings. Pre-swap checklist (Lab 5): confirm partner PSU is healthy **first**.

Step 6 — CMOS battery failure

Server+ 4.2 calls this out. Symptoms:

- Clock skew on every boot (``date`` jumps wrong) — Lab 31 callback.
- BIOS settings reset to defaults.
- Boot order forgets the OS disk.

```
date
chronyc tracking 2>/dev/null | head
```

Fix: replace CR2032 battery; re-set BIOS settings; verify NTP sync.

Step 7 — POST errors and visual / auditory cues

Server+ explicitly tests **POST codes, LED, LCD, auditory, olfactory** (yes — burnt smell is a clue).

Cue	Likely cause
Long beep, repeating	Memory error
3 short beeps	Video card / GPU
No POST + amber LED	PSU / power good signal
LCD: CPU thermal trip	CPU fan failure / heat sink loose
Burnt smell	Capacitor / VRM failure → power down immediately
Clicking HDD	Head crash imminent → replace

Step 8 — Cards, cables, and "reseating"

```
lspci -tv | head
lshw -short -class storage -class network 2>/dev/null | head
```

Field fix list for intermittent device drop-outs:

1. Power off; ESD strap.
2. Remove and ****reseat**** PCIe card and any SAS / SATA cables.
3. Inspect for bent pins, dust, corrosion.
4. ****Compressed air**** at the slot; don't use a vacuum (static).
5. Power on, re-check ``lspci`` / ``dmesg``.

Step 9 — Environmental causes

Factor	Effect on hardware	Mitigation
Dust	Insulates components → over-temp	Schedule cleaning
Humidity high	Condensation, corrosion	HVAC dehumidify (Lab 19)
Humidity low	Static discharge	HVAC humidify, ESD wrist straps
Temperature	Thermal throttle, drive wear, capacitor degradation	ASHRAE 18–27 °C

Step 10 — Misallocated virtual resources (VM tie-in)

For a VM that's "slow" with no obvious hardware fault:

```
top -bn1 | head
free -h
iostat -x 1 3
```

Common causes:

- vCPU over-subscribed (Lab 15) — host CPU at 100 %.
- Memory ballooning / swap — guest swapping is the tell.
- Storage over-subscribed — high `await`.

Resolution: move the VM or rebalance the host.

Step 11 — Toolkit you'll carry to the rack

Tool	Purpose
ESD wrist strap	Prevent component damage
Compressed air	Dust removal
Multimeter	PSU voltage check
Spare DIMMs / PSUs / drives	Hot-spare swap
USB-serial cable	Console access when LAN dies
Live USB diag image	`memtest86+`, vendor diagnostic ISO

Step 12 — Decision tree

```
Symptom → check event logs first
          → no clue → check sensors (temp, fan, voltage)
                    → over-temp → check fans, heat sink, dust
                    → voltage fault → check PSU
          → memory log entries → run memtest86+
          → drive symptoms → smartctl -H ; replace if predictive failure
          → still unclear → reseal cards/cables → vendor diagnostic ISO
```

Free online tools

- **smartmontools** — <https://www.smartmontools.org/>
- **memtest86+** — <https://www.memtest.org/>
- **EDAC docs** — <https://www.kernel.org/doc/html/latest/driver-api/edac.html>

What you learned

- Event-log-first investigation discipline.

- Mapping each Server+ 4.2 symptom (BSOD/PSOD/dump, kernel panic, lockups, POST codes, CMOS battery, predictive failures, cooling) to a concrete tool.
- SMART attributes that predict drive failure.
- BMC sensor / SEL workflow for PSU and fan faults.
- Environmental causes and ESD discipline at the rack.
- Distinguishing host-hardware faults from misallocated-VM symptoms.